Identifying the **Context Shift** between Test Benchmarks and Production Data

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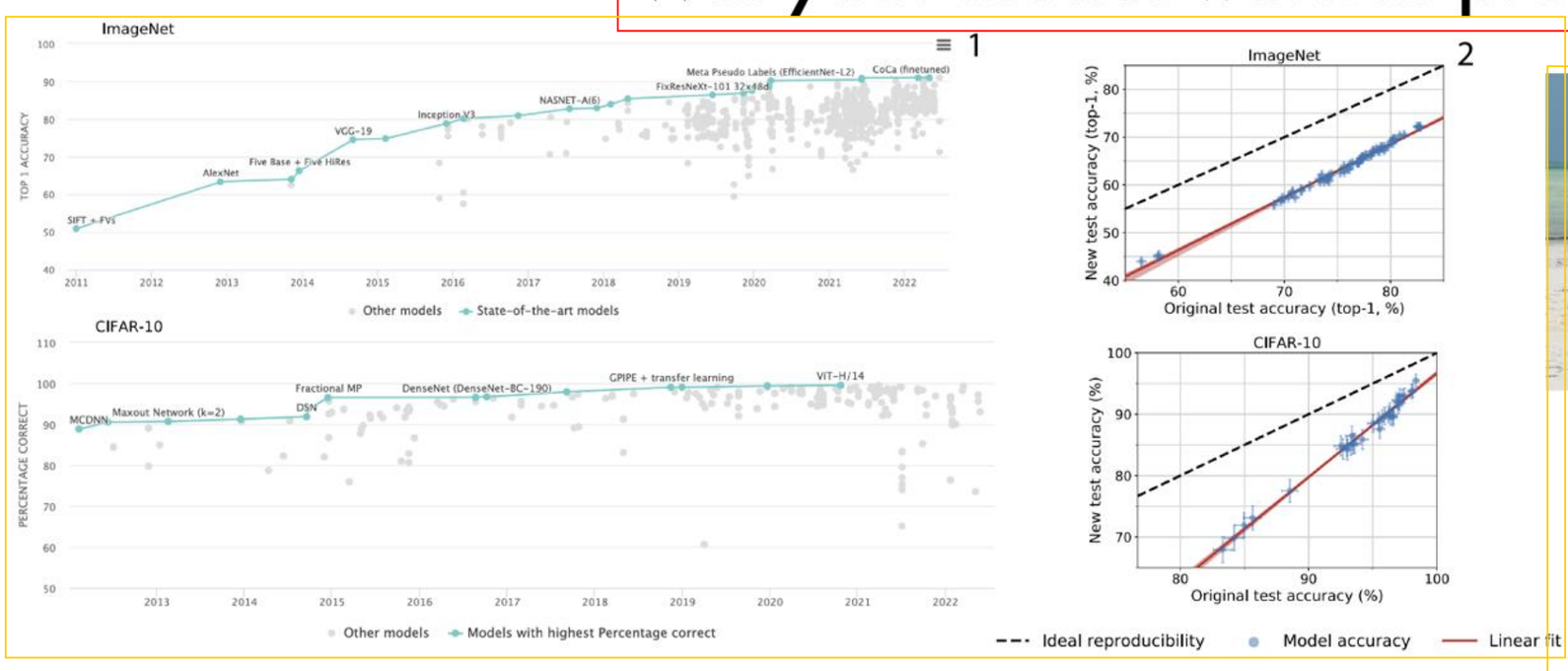
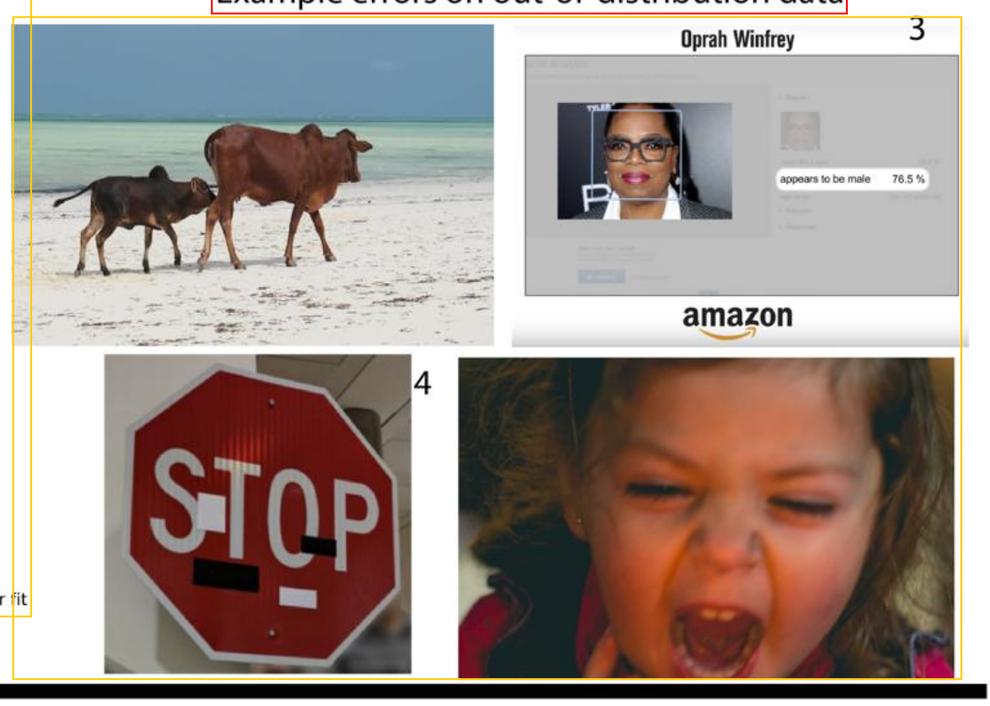
Will your model work in production?



Example errors on out-of-distribution data



Consider the Data Generating Process

Distribution Shift Perspective

Covariate Shift: $P_{test}(x) \neq P_{train}(x)$ and $P_{test}(y|x) = P_{train}(y|x)$

Prior Probability Shift: $P_{test}(y) \neq P_{train}(y)$ and $P_{test}(y|x) = P_{train}(y|x)$

Concept Shift: $P_{test}(x) = P_{train}(x)$ and $P_{test}(y|x) \neq P_{train}(y|x)$

Other Distribution Shift: $P_{test}(x,y) \neq P_{train}(x,y)$

Context Shift Perspective

Sample Selection Bias: when the data distribution is missing relevant examples or dimensions

Non-stationarity: when the data distribution changes over time or environment

Adversarial Perturbations: when the distribution of data is altered in a way that does not affect human task performance

Contextual Dimensions in Human Centered ML Applications

skin color in face recognition (Buolamwini & Gebru 2018) and dermatology diagnosis (Groh et al 2021, Daneshjou et al, 2022), background scenery for affect recognition (Kosti et al, 2019), number of people in a video for deepfake detection (Groh et al 2022), number of chronic illnesses for algorithmic healthcare risk prediction (Obermeyer et al, 2019), data artifacts like surgical markings (Winkler et al 2019) or clinically irrelevant labels (Oakden-Rayner et al 2020) for medical diagnosis classification, and patients' demographics and self reports of pain for quantifying severity of knee osteoarthritis (Pierson et al, 2021)

Who is represented in the data? Who is annotating the data? When and where is data collected?

How do social, geographical, temporal, aesthetic, financial, and other idiosyncrasies influence the data?

Data Generation Process Desiderata for Dynamic Benchmarks

1. **Prediction Task**: What are the input features and output labels?
2. **Test Size**: What is the minimum test size for reducing within data sampling error to an acceptably small rate?
3. **Ground Truth Annotation Arbitration**: Who has the authority to annotate the data? How should disagreement be represented?
4. **Data Inclusion and Exclusion Criteria**: What are the possible data sources? How is data curated? What are the quality constraints?

e.g. Dynabench[13]

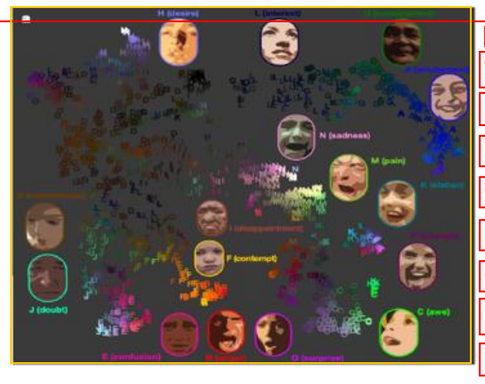
Case Study of Implicit Assumptions in Facial Expression Recognition

Example Performance Evaluation of FER from Mollahosseini et al 2016

	Proposed Architecture	AlexNet
MultiPie	94.7	94.8
MMI	77.9	56.0
DISFA	55.0	56.1
FERA	76.7	77.4
SFEW	47.7	48.6
CK+	93.2	92.2
FER2013	66.4	61.1

How Should Facial Expressions be Represented?

As the following "basic emotion categories:" Neutral, Happiness, Sadness, Anger, Fear, Disgust, Surprise? Or as the emotions on the right from Cowen et al 2020? Or something else?



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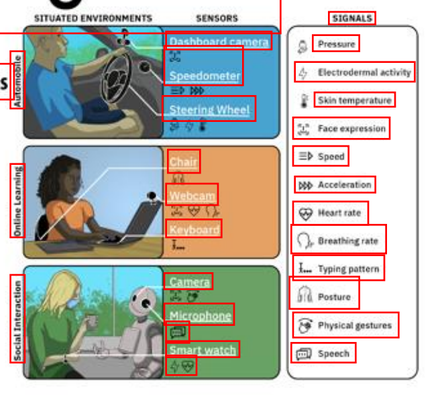
Too many degrees of freedom + too many assumptions

What data can serve as ground truth: posed photographs with observers' labels, frames from a video with observers' labeled peak expressions, self reports, mappings of facial action units to categories? When should we expect the model to generalize? What limits should be express?



References

1. paperswithcode.com/sota/image-classification-on-cifar-10 and paperswithcode.com/sota/image-classification-on-imagenet
2. Recht, et al. Do ImageNet Classifiers Generalize to ImageNet. 2019.
3. Buolamwini and Gebru. Gender Shades: Intersectional Accuracy Diparities in Commercial Gender Classification. 2018
4. Eykholt K, et al. Physicial-World Attacks on Deep Learning Visual Classification. 2018
5. Groh et al. Evaluating Deep Neural Networks Trained on... 2021
6. Daneshjou et al. Disparities in Dermatology AI... 2022
7. Kosti et al. Context Based Emotion Recognition using... 2019
8. Groh et al. Deepfake Detection by Crowds, Machines, and... 2022
9. Obermeyer et al. Dissecting Racial Bias in an Algorithm... 2019
10. Winkler et al. Association Between Surgical Skin Markings in Dermoscopic... 2019
11. Oakden-Rayner et al. Hidden Stratification Causes Clinically Meaninful... 2020
12. Pierson et al. An Algorithmic Approach to Reducing Unexplained... 2021
13. Kiela et al. Dynabench: Rethinking Benchmarking in NLP. 2021
14. Mollahosseini et al. Going Deeping in Facial Expression Recognition... 2016
15. Cowen et al. Sixteen Facial Expressions Occur in Similar Contexts... 2020

Link to Preprint

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